ECA1003 - MPMC

Exp-01

1) P-A

8- bit addition and Subtraction

A = 5AH , B = 27H

Addition :

5AH + 27H = 81H

Sum = 81H, Cy = 0

AC = 1 , P = 1, Z = 0 , S = 1

Flag byte = 94H

Signed Overflow = Yes.

Subtraction :

5AH - 27H = 33H

2's of 27H = 33H

Borrow = 0, AC = 0

P = 1 , Z = 0, S = 0

Flag byte = 06H

Signed Overflow = NO.

Program B

16-bit addition & Subtraction

Gn

X = 1234H , Y = 4321H

Add:

1234H + 4321H = 5511H

Carry = 0.

Sub:

~~1234 - 4321H = CF13H~~

1234 - 4321H = CF13H

Borrow = 1

Program - C:

8- bit Multiply.

14H X 06H

Division F2H ÷ 0AH

Multiply:

14H X 06H

= 0678H

Division:

F2H ÷ 0AH

Quo = 18H

Rem = 02H.

Program - D

16- b

Gn

Multiply

Division

Multiply:

135

=

Upper

Division.

9A 23H

Quo

Rem

Exper

Program-D

16- bit Multiplication & division :-

Gn

Multiply 135FH x 0025H

Division: 9A23H ÷ 0011H.

Multiply :

135F H x 0025H

= 002CA13H

Upper word = 0002H ; Lower word = CA13H.

Division.

9A23H ÷ 0011H

QUO = 0911H

Rem = 0002H.

Expression:

1110 0101 = X

1111 0000 = Mask

1110 0000